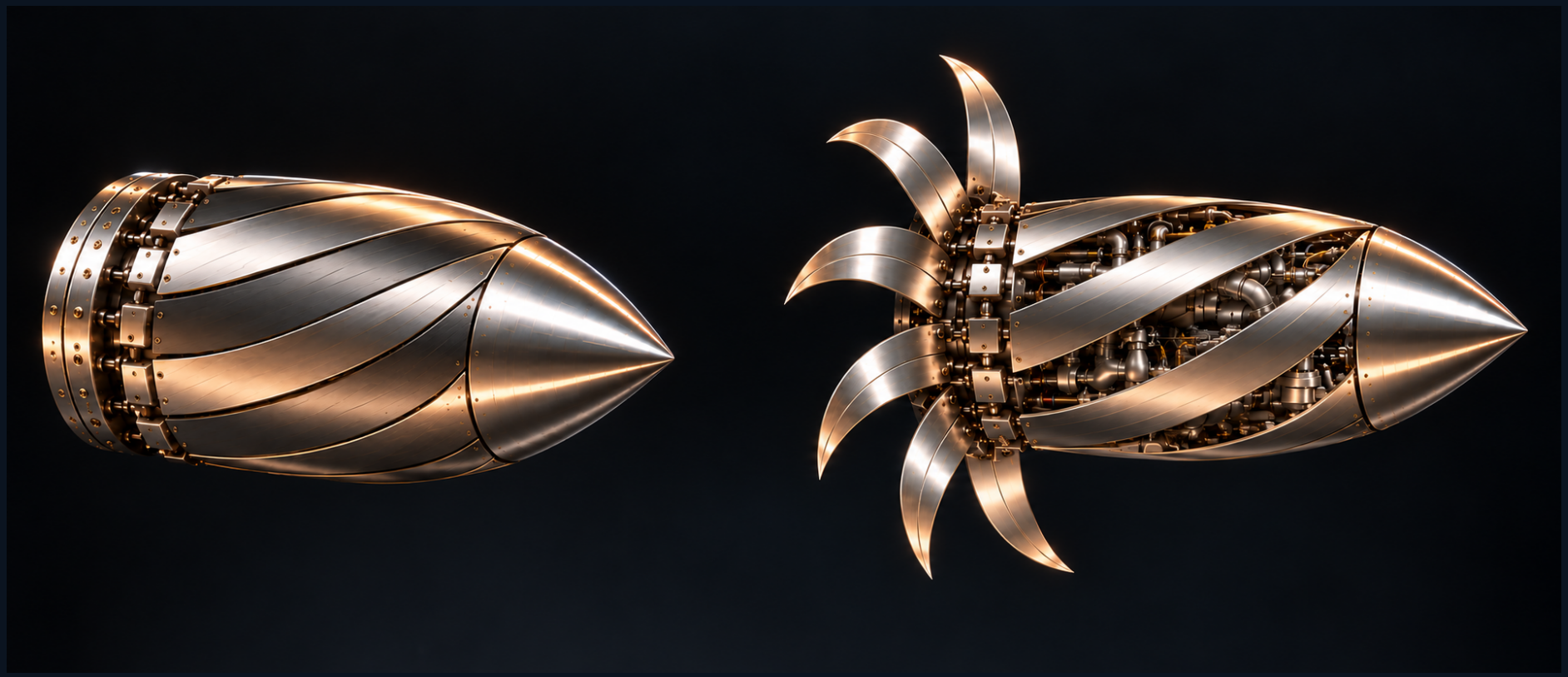
# Engineering portfolio

# Nickolas Yukhnevich

Electrical systems. Embedded control. Electromechanical integration.



INHALER / CLOSED LEFT - OPEN RIGHT

Inhaler and Aetherion: independent electromechanical projects exploring deployment geometry, sensor feedback, coordinated actuation, and power management.

# Portfolio introduction

I am an Arizona State University student specializing in electrical systems concepts, embedded control, and electromechanical integration. This portfolio presents my independent projects, Inhaler and Aetherion, with a focus on sensor feedback, control logic, actuator interfaces, and power management.

My project studies explore how electrical architecture and coordinated actuation support deployable mechanisms. Through configuration views, system diagrams, and technical explanations, each case study communicates the design approach and the relationship between electrical control and mechanical behavior.

**Arizona State University**
University student

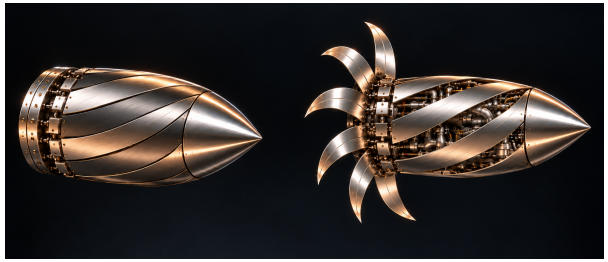
**3.86 cumulative GPA**
Academic performance

**U.S. citizen**
Engineering internship applicant

**Electromechanical systems**
Embedded control & sensor integration

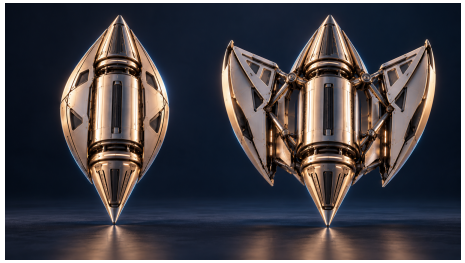
## Inhaler

Deployable-blade capsule / open-shell configuration



## Aetherion

Folding-wing capsule / symmetric transformation



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Electrical engineering internships

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# Inhaler

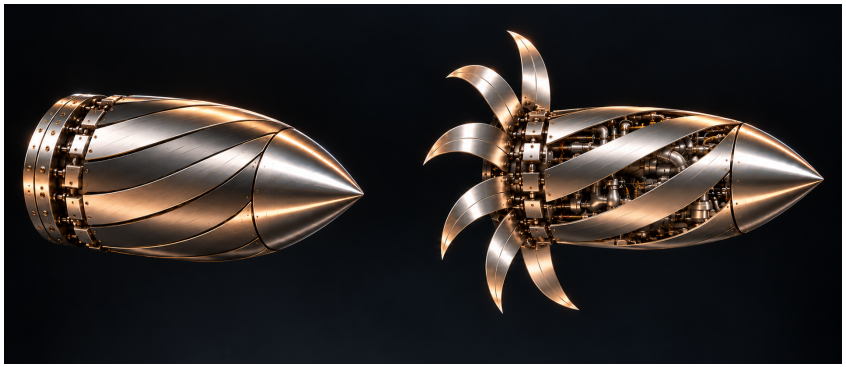
A deployable-blade capsule with an exposed internal compartment in the open configuration.

Inhaler features eight blade panels designed to stow along a capsule and rotate outward, opening shell apertures that reveal the engine compartment between fixed body strips. The design provides a foundation for physical implementation through integrated hinges, actuators, position sensors, and an embedded controller, with prototyping and testing to verify reliable operation.

## Design intent

The compact configuration alternates deployable blade panels with fixed body strips. Opening the blades uncovers their original shell apertures: these gaps remain open and reveal the internal engine compartment, rather than closing behind the moving panels.

My contribution at this stage is the mechanism design: refining the repeated blade shape, the spacing between elements, and the visual relationship between the closed capsule and the open arrangement.



01

### Alternating fixed and moving surfaces

Fixed body strips remain between neighboring blade panels. The target body opening follows the curved outline, sweep, and tapered tip of its blade panel while stowed, revealing the engine compartment when the blade deploys.

02

### Rear-mounted hinges

Each blade is intended to pivot near the rear ring through a nominal 0°-90° travel.

03

### Consistent repeated geometry

The target array uses eight identical blade shapes at 45° intervals, with equal radial reach and tips on a common circle. At full deployment, the array lies in a plane perpendicular to the capsule axis.

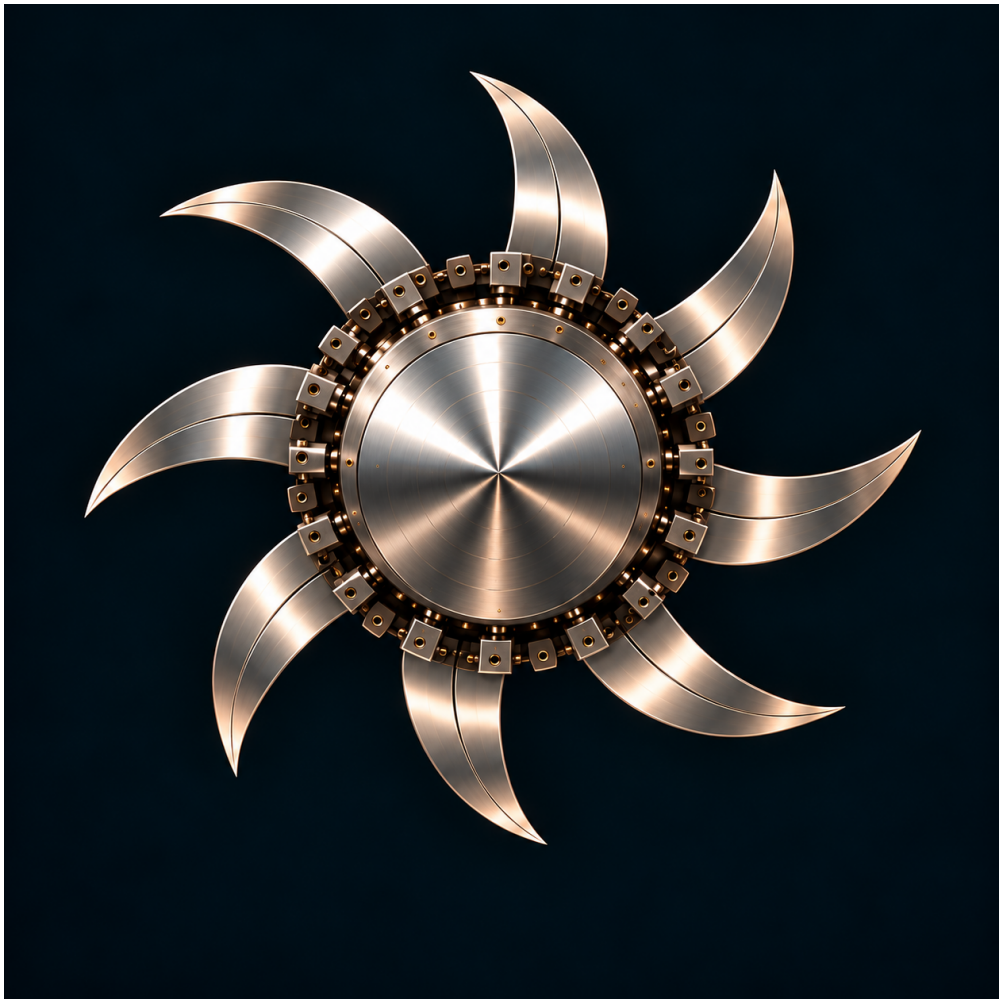
04

### Compact stowage envelope

Folded panels follow the intended body profile; a dimensional CAD model must check panel thickness, hinge intrusion, tip clearance, and tolerance accumulation.

# Inhaler - radial blade geometry

One intended blade profile repeated around the rear ring at equal angular intervals.



## Repeated blade geometry

Each intended blade uses the same swept profile, chord, and radial length. The technical drawing repeats one definition eight times.

## Equal angular spacing

Blade positions are distributed at 45° intervals around the rear ring, with their tips on one circular envelope.

## One deployment plane

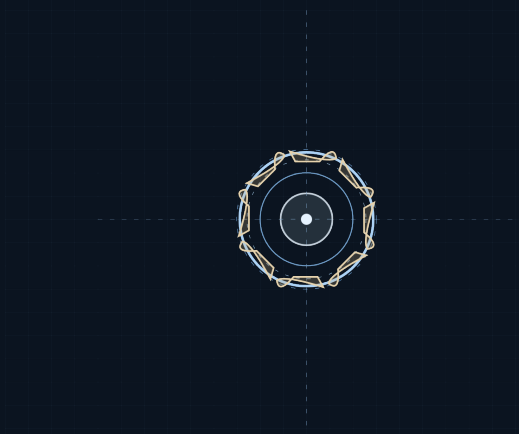
The fully opened array lies perpendicular to the longitudinal body axis. A side view changes the projection, not the lower blade shapes.

## Open body apertures

The target aperture follows its corresponding blade’s stowed outline. The fixed strips remain between the openings, revealing the internal compartment when the panels deploy.

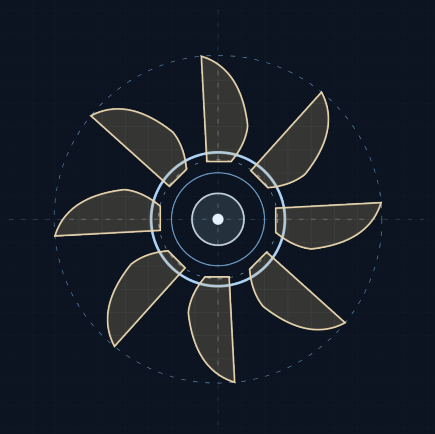
# Inhaler - deployment sequence

The three views illustrate the stowed, intermediate, and fully deployed positions. The schematic highlights uniform blade spacing and equal radial reach, while the side blueprint illustrates the shell openings that expose the internal compartment.



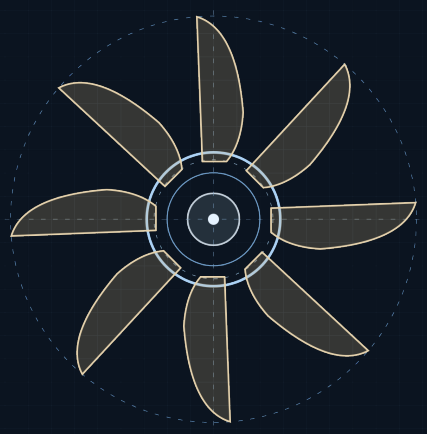
Rear-ring projection / 8 identical blade positions
Dashed circle: common blade-tip envelope

0 DEGREES / STOWED



Rear-ring projection / 8 identical blade positions
Dashed circle: common blade-tip envelope

45 DEGREES / INTERMEDIATE



Rear-ring projection / 8 identical blade positions
Dashed circle: common blade-tip envelope

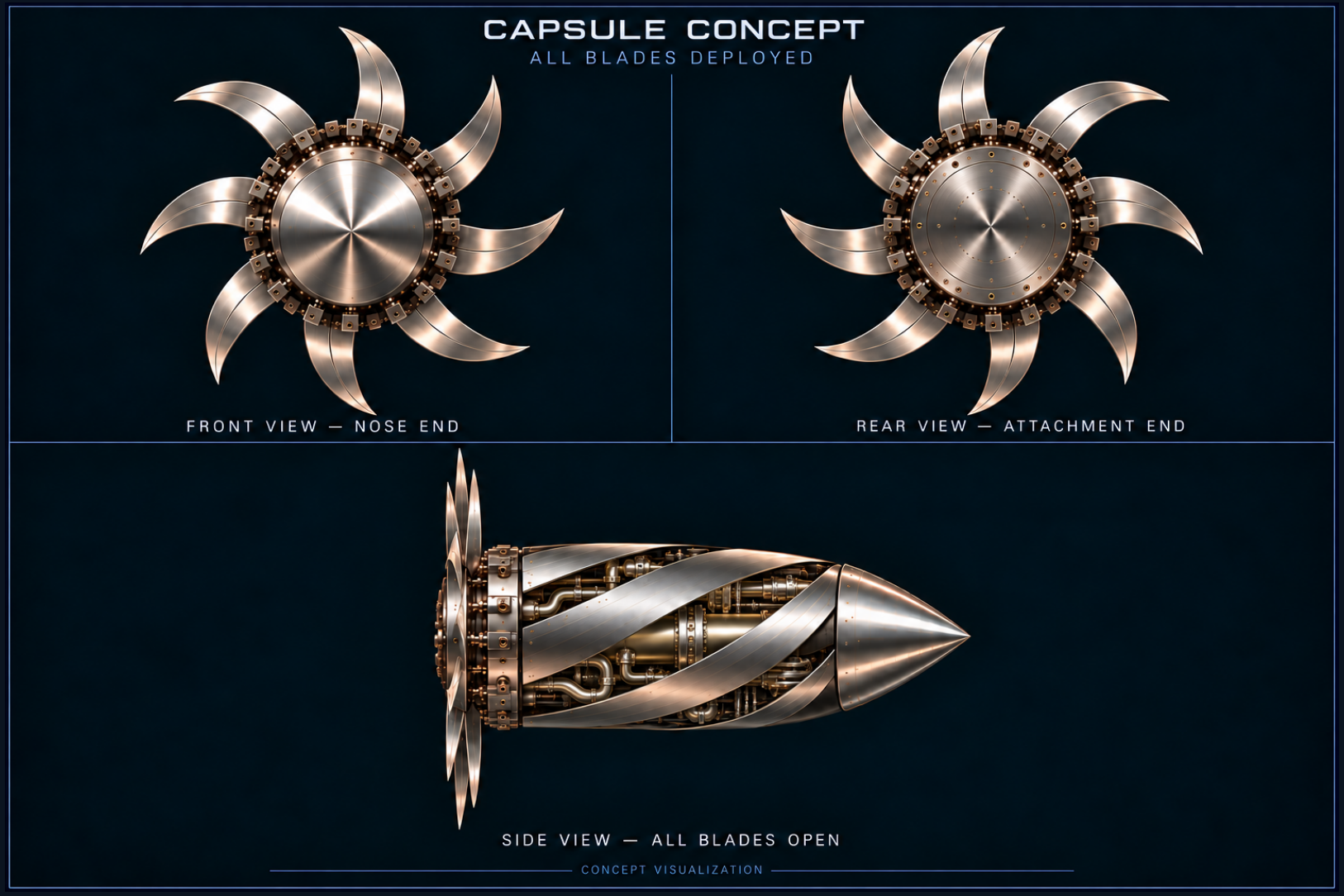
90 DEGREES / DEPLOYED

## Geometry requirements

Eight identical blade profiles; 45-degree angular pitch; equal radial reach; a common blade-tip envelope at full deployment.

# Inhaler - front, rear + side views

All blades deployed / illustrative three-dimensional engine compartment.



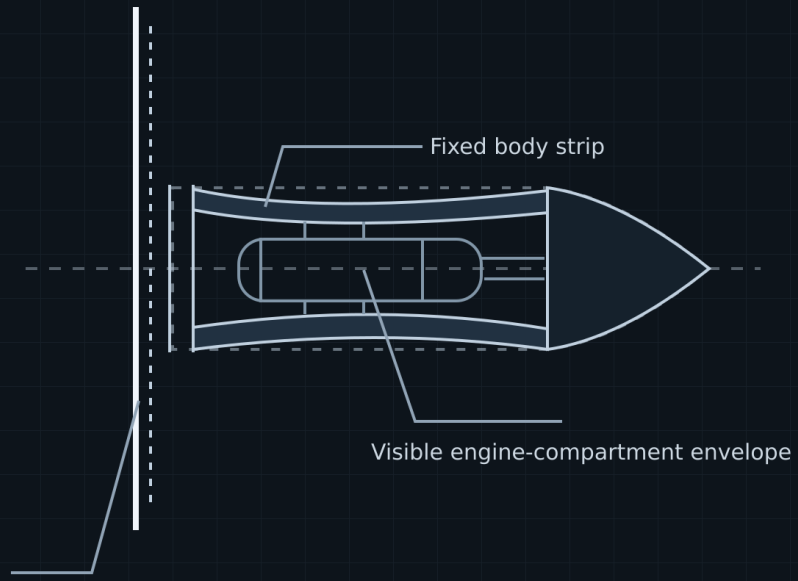
# Inhaler - deployment blueprint

Open shell apertures, internal compartment, rear hinge ring, and eight equally spaced blade positions.

## INHALER / OPEN-SHELL CONFIGURATION

SIDE ELEVATION + REAR PROJECTION

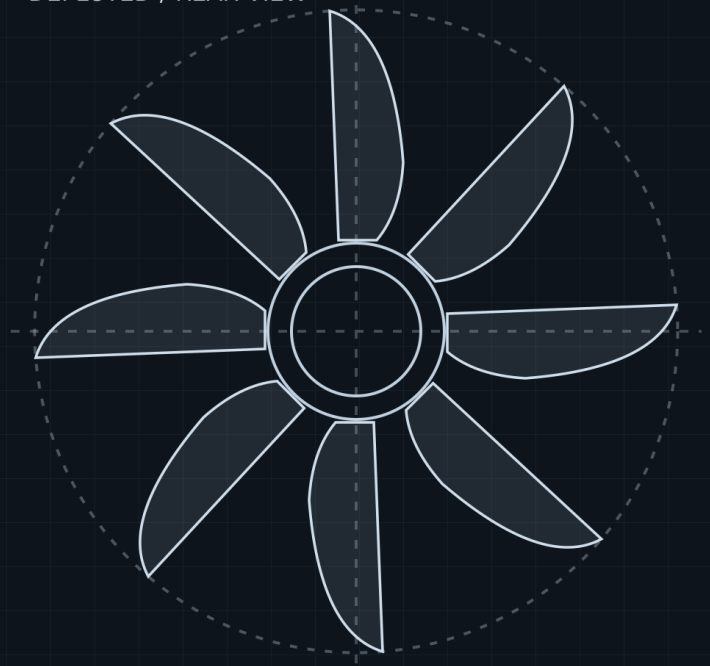
DEPLOYED / TRUE SIDE VIEW




Radial blades viewed edge-on

All blade tips share one circular path.

DEPLOYED / REAR VIEW



8 × identical blade geometry

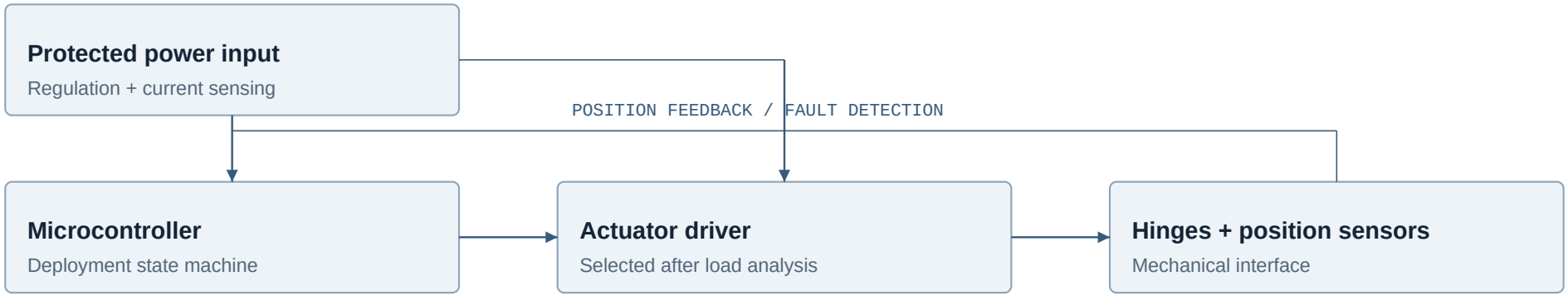
45° equal angular pitch / 90° intended opening

Open apertures remain between fixed strips after blade deployment.

CONCEPT CONFIGURATION · ILLUSTRATIVE INTERNAL VOLUME · NO FABRICATION DIMENSIONS

# Electrical + mechanical integration

The next phase would connect commanded movement to measured position. Component selection should follow mechanical load estimates and power requirements, rather than precede them.




## Mechanical engineering focus

### Parametric CAD and interference study

Build the body, rear ring, and repeated blade assembly in SolidWorks; sweep the hinge motion and check adjacent-panel clearance before producing prototype parts.

### Hinge and actuation sizing

Estimate torque from geometry, friction, and representative external loading; compare independent actuators with a shared linkage before committing to hardware.

## Electrical engineering focus

### Proposed embedded control

Evaluate an ESP32 development board for an initial C++ state machine with stowed, deploying, open, retracting, and fault states; confirm I/O and driver compatibility during selection.

### Proposed circuit development

Use KiCad for a future schematic and two-layer prototype PCB after bench validation, with regulated power, actuator-driver interfaces, position-feedback inputs, and current monitoring.

Tool status: SolidWorks, MATLAB, ESP32, and KiCad describe the proposed engineering workflow. This case study does not claim a completed CAD assembly, manufactured PCB, or working embedded controller.

# Development roadmap

The sequence begins with the current configuration work and identifies the engineering evidence needed before the design can progress toward a functioning assembly.

01 / CONCEPT ESTABLISHED

### Define the compact and deployed configurations

Establish the alternating fixed-panel/blade arrangement, rear-hinge concept, repeated geometry, and the intended transition between closed and open states.

02 / VISUAL STUDY

### Refine the visual design

Compare side and end views to communicate blade spacing and the deployed arrangement; use concept renderings to surface geometry questions for the CAD phase.

03 / PLANNED

### Create a parametric assembly

Set dimensions, material thicknesses, hinge axes, stops, and tolerances in CAD; use a motion study to identify collisions and inaccessible assembly features.

04 / PLANNED

### Prototype one hinge and blade

Build a single-element test fixture to measure travel, friction, play, and required torque before duplicating the mechanism around the full body.

05 / PLANNED

### Integrate the control and power system

Connect the selected actuator and position sensor on a development board; establish current limits, end-stop handling, and a safe response to missing feedback.

06 / PLANNED

### Validate repeated deployment

Test the assembled mechanism across repeated cycles and record position consistency, cycle time, current draw, temperature, wear, and fault behavior.

# Verification plan

A successful rendering does not establish mechanical or electrical performance. These proposed checks would produce the evidence needed to revise the design and report defensible results.

| Engineering question | Proposed method | Evidence to record |
| --- | --- | --- |
| **Clearance across travel** | Sweep the CAD assembly and inspect a slow-moving physical fixture. | Minimum separation, interference locations, and changes required. |
| **Position repeatability** | Record blade angle over repeated opening and closing cycles. | Angle spread, end-stop consistency, and measured backlash. |
| **Actuator load and power** | Measure current, voltage, and motion under defined bench loads. | Peak current, steady current, torque estimate, and power margin. |
| **Fault response** | Interrupt sensor feedback or obstruct motion under controlled bench conditions. | Detection behavior, output shutdown, recovery sequence, and retained faults. |
| **Environment and materials** | Define operating conditions before selecting seals and conducting exposure tests. | Material compatibility, enclosure requirements, and post-test inspection. |

Open design question: The visible internal machinery communicates an engine-compartment envelope. Engine construction, performance, airflow, loading, and operation remain outside the current mechanism design.

# Aetherion

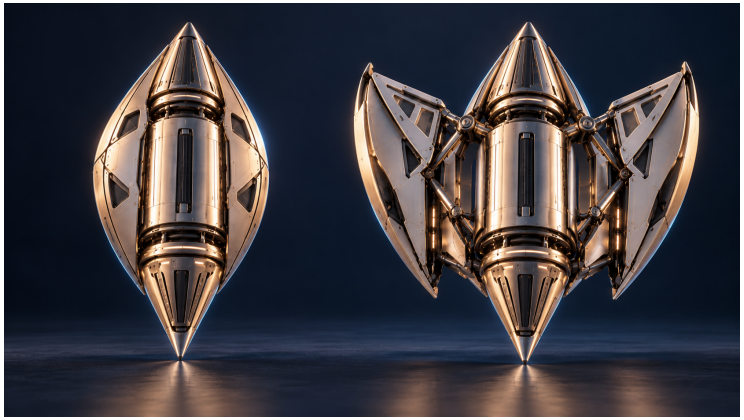
A transforming capsule concept with four symmetrically deployable wings.

Aetherion explores a transition from a compact capsule to a four-wing configuration. The visual study centers on consistent wing shape, symmetric deployment, and a sequence that could later be implemented through mechanical linkages, position sensing, and embedded control.

## Design intent

Aetherion is a transforming capsule concept featuring four symmetrically arranged folding wings. Its design emphasizes compact packaging, consistent wing geometry, clearance around the wing roots, and retention of the deployed configuration.

The project integrates these mechanical considerations with a proposed electrical architecture for position sensing, coordinated actuation, and deployment-state monitoring. Configuration drawings, system diagrams, and a transformation storyboard illustrate how sensor feedback and control logic would support consistent wing movement and identify incomplete or uneven deployment.



01

### Four repeated wing elements

The visual arrangement uses four symmetrically distributed wing petals around the capsule; equal shape and spacing are the starting geometry requirements.

02

### Compact-to-open transition

The closed concept brings the wing surfaces close to the body, while the deployed concept separates them into a wider configuration.

03

### Controlled motion and retention

The engineering phase must define hinge axes, clearance, hard stops, latching, and the means of retaining each configuration.

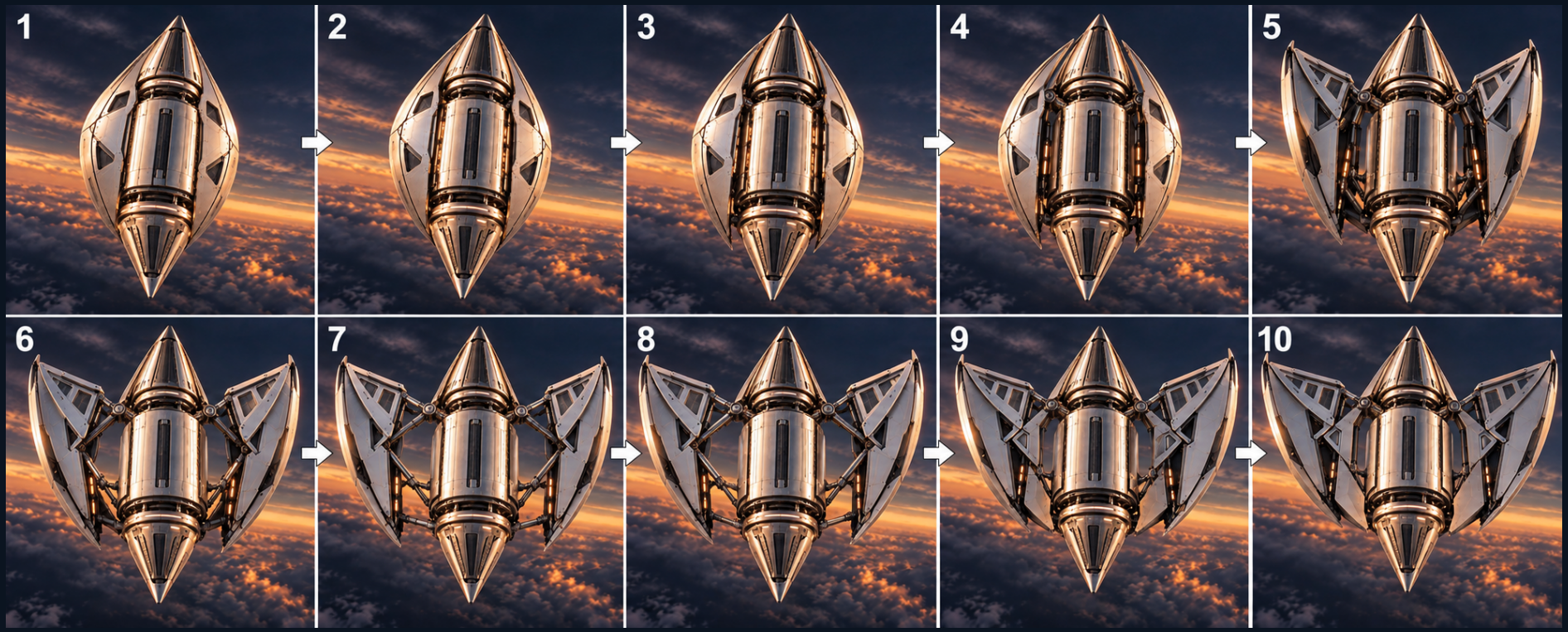
04

### Balanced deployment

The proposed control logic compares wing position feedback to detect incomplete or uneven movement and confirms deployment only when all wings reach their designated positions.

# Aetherion - transformation storyboard

The intended transition from a compact capsule to an open folding-wing configuration.




The concept storyboard illustrates the transition from compact stowage to the intended deployed configuration, highlighting wing alignment, symmetry, and the opening sequence.

# Aetherion - deployment questions

Engineering questions associated with each frame of the visual storyboard.

01

### Compact configuration

The capsule establishes the stowed envelope; a CAD model would define how wing thickness and hinge packaging fit inside this profile.

02

### Stowed wing alignment

The repeated panels remain near the body; the engineering task is to establish consistent closed-state alignment and tolerances.

03

### Ready-to-deploy configuration

The compact form is maintained before opening; a future control sequence would verify sensor state and release conditions.

04

### Initial wing separation

The first visible opening introduces clearance questions around the wing roots and neighboring surfaces.

05

### Intermediate travel

Wing elements move farther from the body; CAD and a fixture would check the swept volume and any interference.

06

### Approaching the open form

The wider configuration appears; the mechanism must approach its stops without impact that exceeds component limits.

07

### Open geometry study

The four-wing arrangement is visible; measure relative wing angles to evaluate symmetry in a future prototype.

08

### Open-state consistency

The repeated open form emphasizes shape consistency; a detailed assembly would specify allowable position variation.

09

### Retention and sensing

The deployed configuration would need mechanical retention and position feedback before the controller reports completion.

10

### Intended deployed configuration

The final storyboard form communicates the design goal; repeated bench tests must establish whether the mechanism reaches and holds it reliably.

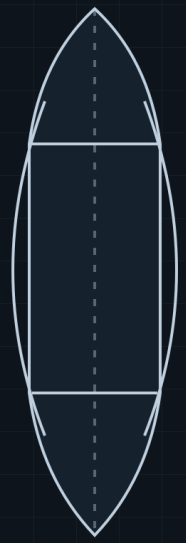
# Aetherion - folding-wing blueprint

Pointed capsule ends, longitudinal folding wings, wing-root interfaces, and four-wing symmetry.

AETHERION / FOLDING-WING CAPSULE

PROJECT-SPECIFIC CONFIGURATION DRAWING

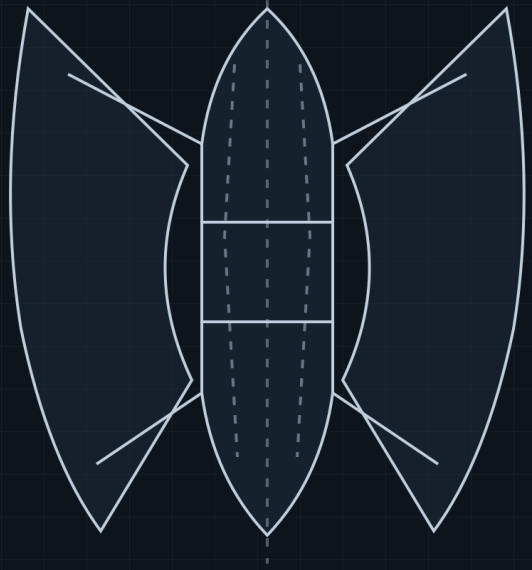
STOWED / FRONT VIEW



Pointed upper and lower ends

Wing surfaces follow the stowed envelope.

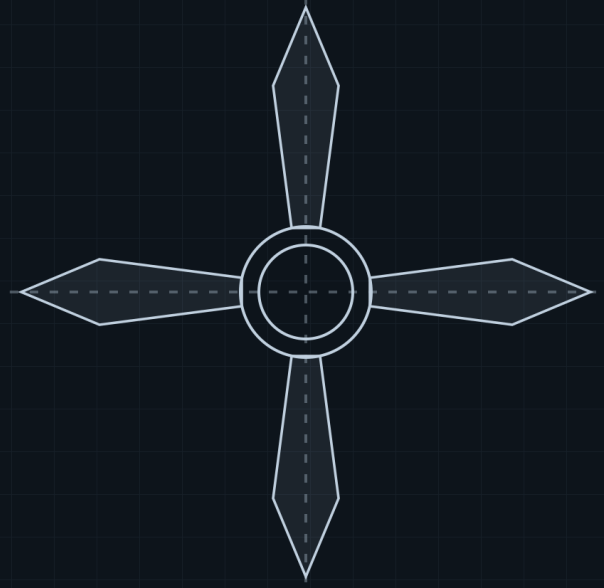
DEPLOYED / FRONT VIEW



Mirrored folding wing panels

Two visible wings; concealed panels indicated.

DEPLOYED / END VIEW



4-wing symmetry / 90° spacing

Axial projection of wing spans.

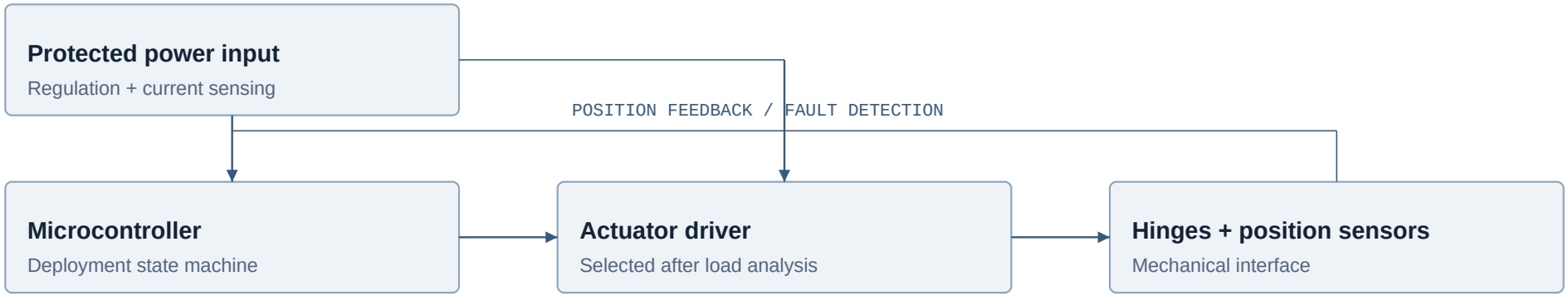
Folding panels extend alongside the capsule; wing outline is distinct from Inhaler blades.

Hinge and linkage locations are illustrative interfaces for further CAD definition.

CONCEPT CONFIGURATION · NOT TO SCALE · NO FLIGHT OR MANUFACTURING VALIDATION

# Electrical + mechanical integration

The proposed electronics would manage a bench deployment mechanism. Power requirements, actuator architecture, and sensing should be selected together with the hinge design.




## Mechanical engineering focus

### Wing-root and linkage geometry

Develop the folding-wing assembly in SolidWorks, checking the complete swept volume, root clearances, and the transition into the open hard-stop position.

### Structural and packaging study

Compare material and hinge concepts against the intended bench loads; use a simplified MATLAB motion model to organize angle, travel, and timing requirements.

## Electrical engineering focus

### Proposed controller and feedback

Evaluate an ESP32-based C++ prototype with separate wing-position inputs and a deployment state machine that distinguishes commanded motion from confirmed position.

### Proposed PCB workflow

Use KiCad for a future two-layer interface board after selecting actuators, with driver connections, regulated logic power, sensor inputs, and accessible bench-test points.

Engineering approach: Proposed software, controller, and PCB architecture for coordinated actuation, sensor feedback, and deployment monitoring.

# Development roadmap

The development plan centers on a controlled bench mechanism to evaluate deployment consistency, sensor feedback, and coordinated actuation.

01 / CONCEPT ESTABLISHED

### Establish the visual concept

Define the compact capsule and the intended four-wing arrangement, focusing on symmetry and consistent repeated geometry.

02 / VISUAL STUDY

### Document the intended sequence

Use the ten-frame storyboard to communicate the transformation and identify the points where clearance, hinge travel, and retention need engineering definition.

03 / ENGINEERING SCOPE

### Model the wing assembly

Create the parametric geometry, define hinge coordinates and stops, and inspect the swept volume throughout the complete opening and closing sequence.

04 / ENGINEERING SCOPE

### Build a single-wing fixture

Measure hinge travel, required actuation force, backlash, and stop behavior on one wing before assembling the complete mechanism.

05 / ENGINEERING SCOPE

### Implement synchronized bench deployment

Integrate the chosen actuator architecture and position sensors; compare feedback against commanded states and handle incomplete deployment explicitly.

06 / ENGINEERING SCOPE

### Record repeatability and fault behavior

Run defined deployment cycles, inspect wear, and document position agreement, timing, current, and controlled fault responses.

# Verification plan

The proposed test plan evaluates unobstructed movement, accurate position sensing, and consistent locking in the deployed configuration.

| Engineering question | Proposed method | Evidence to record |
| --- | --- | --- |
| **Wing clearance** | Inspect the CAD swept volume and slowly cycle a physical fixture. | Interference locations, clearance measurements, and geometry revisions. |
| **Deployment symmetry** | Measure each wing angle across repeated bench cycles. | Relative angle differences, timing variation, and root backlash. |
| **Open-state retention** | Apply defined bench loads to the stopped or latched assembly. | Deflection, latch behavior, and any permanent deformation. |
| **Electrical behavior** | Measure actuator current and sensor outputs throughout travel. | Peak current, supply variation, position consistency, and thermal observations. |
| **Deployment Verification** | Introduce a controlled obstruction or disconnect a position signal. | Fault recognition, shutdown behavior, status reporting, and recovery steps. |

Verification priorities: movement clearance, position feedback, retention, electrical behavior, and controlled fault response.